

Q

Digital Differentiator

Ideal Diff. has a frequency response - linearly proportional to frequency.

Freq. response :

$$H_d(e^{j\omega}) = j\omega, \quad |\omega| \leq \pi.$$

$$= \begin{cases} j\omega & , 0 \leq \omega \leq \pi \\ -j\omega & , -\pi \leq \omega < 0 \end{cases}$$

Also

$$H_d(e^{j\omega}) = \begin{cases} \omega \cdot e^{j\pi/2}, & 0 \leq \omega \leq \pi \\ \omega \cdot e^{-j\pi/2}, & -\pi \leq \omega < 0 \end{cases}$$

$$H_d(e^{j\omega}) = |H_d(e^{j\omega})| \cdot e^{j\angle H_d(e^{j\omega})}$$

For positive frequencies, phase = $\pi/2$.

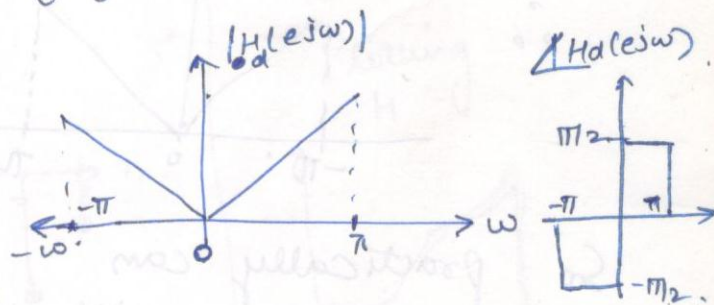
For negative " , phase = $-\pi/2$.

Thus, differentiator shifts the phase $\cos \xrightarrow{j\omega} \sin$.

$$\Rightarrow H_d(e^{j\omega}) = j\omega = |\omega| e^{j\frac{\pi}{2}\text{sign}(\omega)}.$$

$$\Rightarrow |H_d(e^{j\omega})| = |\omega| = \begin{cases} \omega, & 0 \leq \omega \leq \pi \\ -\omega, & -\pi \leq \omega < 0 \end{cases}$$

$$\angle H_d(e^{j\omega}) = \begin{cases} +\pi/2, & 0 \leq \omega \leq \pi \\ -\pi/2, & -\pi \leq \omega < 0 \end{cases}$$



This desired freq. response of diff. is always fixed.

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} \cdot d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} j\omega e^{j\omega n} \cdot d\omega$$

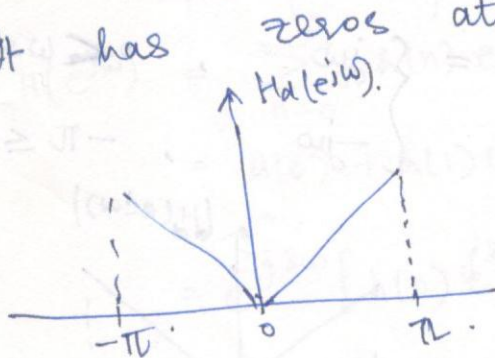
For $n=0$, we
can't solve
as then $\frac{j\omega n}{n}$
can't be solved.
So taking 2 cases.

$$h_d(n) = \begin{cases} 0, & n=0 \\ \frac{\cosh(n\pi)}{n}, & n \neq 0 \end{cases}$$

(Anticausal)

M: odd, M-1: even

If we design diff. with Type -3.
It has zeros at $z=1, z=-1$
ie. $\omega=0, \omega=\pi$.

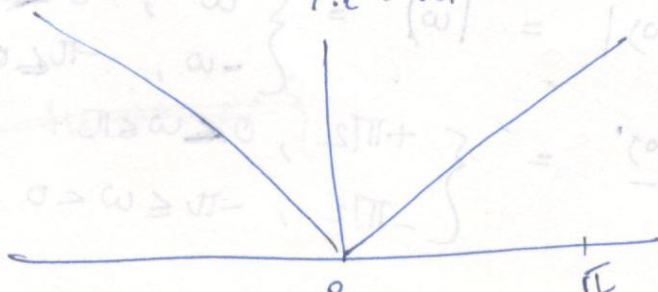


So practically can
be used for diff.
low frequencies.

So ~~the~~ full band
differentiator cannot
design. It can
be designed until its
linear range.

But with Type -4, most suitable to design
diff. as this constraint doesn't apply.

It has zeros $z=1$, not at $z=-1$.
ie. at $\omega=0$.



$$h(n) = -n(M-1-n)$$

$$H(z) = -z^{-(M-1)} H(z^{-1})$$

$$H(1) = -H(1) = 0$$

$$H(-1) = H(-1) \neq 0$$

M: even

Exo.

Given ~~causal~~

desired freq. resp. of differentiator
(will be delayed by τ)

$$\Rightarrow H_d(e^{j\omega}) = j\omega e^{-j\omega\tau}, \quad |\omega| \leq \pi$$

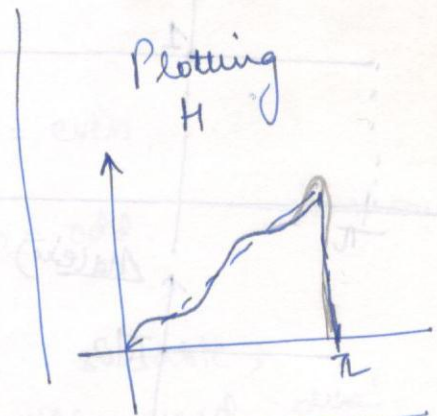
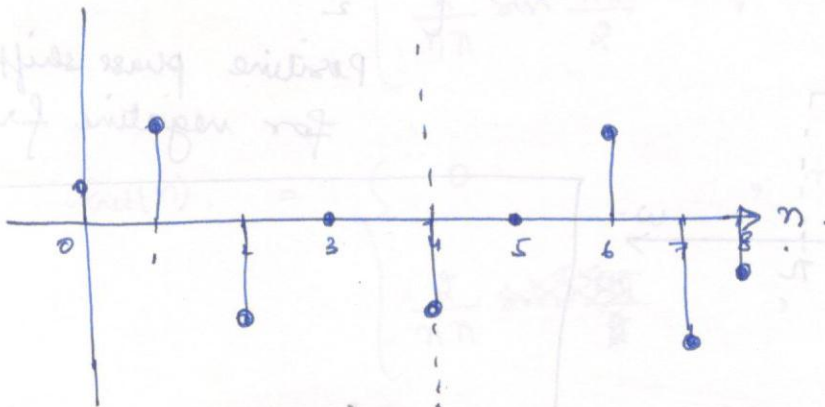
Designing FIR diff using Hamming window, $M=9$. {Type: 3}

$$\tau = \frac{M-1}{2} = 4$$

General exp: $h_d(n) = \begin{cases} 0, & n = \tau \\ \frac{\cos \pi(n-\tau)}{(n-\tau)} - \frac{\sin \pi(n-\tau)}{\pi(n-\tau)}, & n \neq \tau \end{cases}$

$\sin(n-4)\pi = 0$

$= \begin{cases} 0, & n = 4 \\ \frac{\cos \pi(n-4)}{(n-4)}, & n \neq 4 \end{cases}$ computing $h_d(n)$ for $0 \leq n \leq 8$,
 $h_d(n) = \left\{ -\frac{1}{4}, \frac{1}{3}, -\frac{1}{2}, 1, 0, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4} \right\}$



For $M=9$, causal Hamming window $\tau=4$,
 $w(n) = \begin{cases} 0.54 - 0.46 \cos(\pi n/4), & 0 \leq n \leq 8 \\ 0, & \text{otherwise} \end{cases}$

$h(n) = h_d(n) \cdot w(n)$

$h(0) = -h(8)$
 $h(1) = -h(7)$
 $h(2) = -h(6)$
 $h(3) = -h(5)$
 $h(4)$

linear phase structure used for realization

That structure is differentiator

Design the same diff. with type -4.

change M

$\Rightarrow w(n) = \{0.08, 0.2147, 0.54, 0.8653, 1, 0.8653, 0.54, 0.2147, 0.08\}$

Freq. response system function of linear phase

$H(e^{j\omega}) = \sum_{n=0}^8 h(n) e^{-j\omega n}$ = expand

$\Rightarrow H(e^{j\omega}) = j\omega [1.7305 \sin \omega]$

DIGITAL HILBERT TRANSFORMER.

(All pass filter) which imparts phase shift of $\pm \pi/2$ on I/P signal. 30/10/14 Thursday

Frequency response of analog Hilbert transformer.

$$H(e^{j\omega}) = -j \text{sign}(\omega) \quad \text{analog freq. } (\infty \text{ range})$$

Digital HT.

$$H(e^{j\omega}) = -j \text{sign}(\omega), \quad |\omega| < \pi.$$

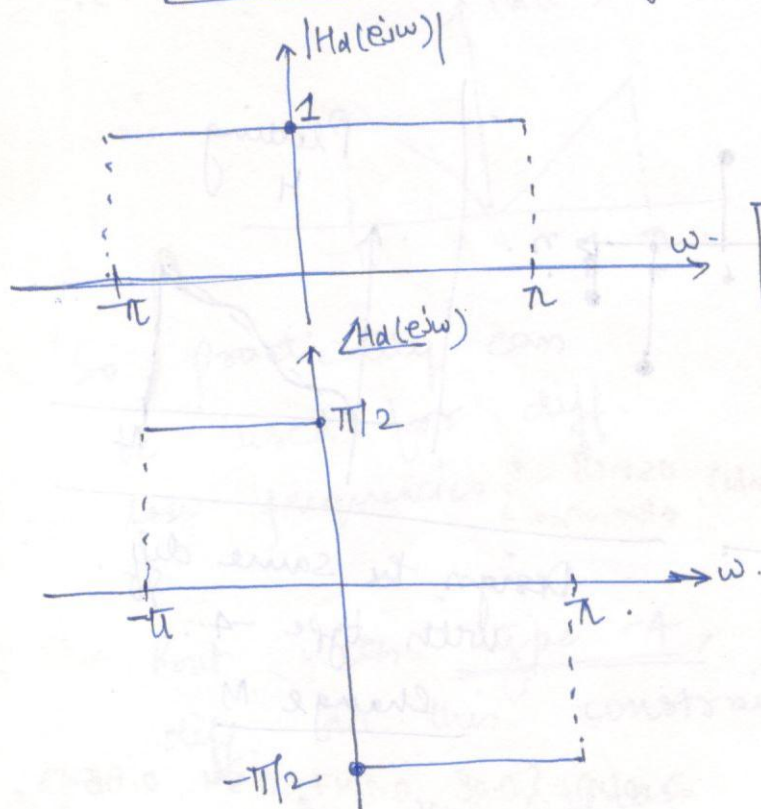
$$= |\text{sign}(\omega)| \cdot e^{j\pi/2 \text{sign}(\omega)}.$$

$$H(e^{j\omega}) = |H_d(e^{j\omega})| \cdot e^{j\angle H_d(e^{j\omega})}$$

$$\Rightarrow |H_d(e^{j\omega})| = |\text{sign}(\omega)| = 1, \quad \forall \omega \text{ or } |\omega| < \pi.$$

$$\angle H_d(e^{j\omega}) = -\frac{\pi}{2} \text{sign}(\omega) = \begin{cases} -\frac{\pi}{2} & 0 < \omega < \pi \\ +\frac{\pi}{2} & -\pi < \omega < 0 \end{cases}$$

Positive phase shift for negative frequencies



(Antisymmetric)

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) \cdot e^{+j\omega n} \cdot d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} -j \operatorname{sign}(\omega) e^{+j\omega n} \cdot d\omega$$

~~$h_d(0)$~~ $h_d(n) \Big|_{n=0} = h_d(0) = 0$ (Integrating above we get zero)

$$h_d(n) = \frac{2}{n\pi} \cdot \sin^2\left(\frac{n\pi}{2}\right), \quad n \neq 0$$

infinite samples.

Designing $h_d(n)$ as FIR, truncate it with window.

$$h_d(n) = \begin{cases} 0 & , n=0 \\ \frac{2}{n\pi} \sin^2 \frac{n\pi}{2} & , n \neq 0 \end{cases} \rightarrow \text{If } n = \text{even, then it } = 0$$

$$h_d(n) = \begin{cases} 0 & , n = \text{even} \\ \frac{2}{n\pi} \sin^2 \frac{n\pi}{2} & , n = \text{odd} \end{cases} : \text{non-causal}$$

Theoretically no type of FIR is suitable, but Type 3 & 4 can be still used due to imaginary phase type.

Use type 3 & 4 with some constraints.

For type-3.

$$H_d(e^{j0}) = 0$$

$$H_d(e^{j\pi}) = 0$$

For type-4.

$$H_d(e^{j0}) = 0$$

but

$$H_d(e^{j\omega}) = 1$$

Applications of HT.

SSB

Secure communication -

2-D HT - image processing.

Analysis of Band pass. ~~filter~~ signals modulators, demodulators.

we apply phase shifting only upto a cut-off frequency.

for causal

$$H_d(e^{j\omega}) = -j \operatorname{sign}(\omega) \cdot e^{-j\tau\omega}$$

$$= \begin{cases} -je^{-j\tau\omega} & , 0 < \omega < \pi \\ je^{j\tau\omega} & , -\pi < \omega < 0 \end{cases}$$

shift by τ , $h_d(n) = \begin{cases} 0 & , n = \tau \\ \frac{2}{(n-\tau) \cdot \pi} \sin^2 \frac{(n-\tau)\pi}{2} & , n \neq \tau \end{cases}$

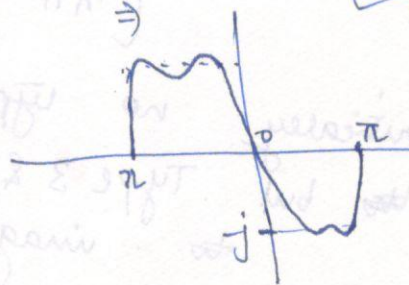
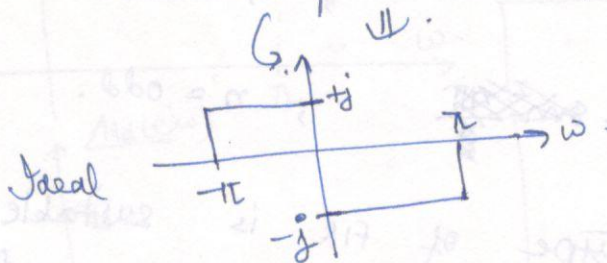
let use hamming window, $M=9$.
 $\tau=4$. $\therefore$ antisymmetric abt 4.

$$h(n) = h_d(n) \cdot w(n), \quad 0 \leq n \leq 8$$

$H(e^{j\omega})$... freq. response of desired HT.

plot its magnitude resp (=1).
 " " amplitude response.

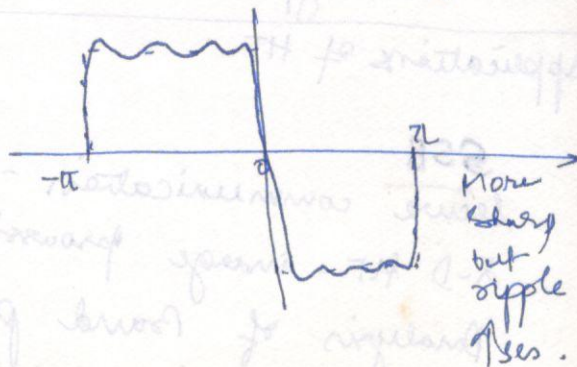
Always plot the amplitude response of HT for more clarity



Look for type 3 as zero at $\omega=0, \pi$.

Can compute $H(z)$.
 Realize $H(z)$ also with linear phase structures.

↓ increasing length



2. Frequency Sampling Method

DTFT
 $\Downarrow$ Sampling
 DFT . get

3. Optimal Method

(Equiripple / Minimax / Parks - McClellan Algorithm / Remez exchange Algo.)
 (Chebyshev Approximation).

For same specifications, length of filter decreases.
 adv, delay, elements, computation, uses.

$H_d(e^{j\omega})$... desired amplitude response

$$H_d(e^{j\omega}) = Q(\omega) P(\omega).$$

$Q(\omega)$	$P(\omega)$
Type I: 1	$\sum_{k=0}^L a(k) \cos k\omega$
II: $\cos \frac{\omega}{2}$	$\sum_{k=0}^L b(k) \cos k\omega$
III: $\sin \omega$	$\sum_{k=0}^L c(k) \cos k\omega$
IV: $\sin \frac{\omega}{2}$	$\sum_{k=0}^L d(k) \cos k\omega$

Difference of desired & designed.

$$\Rightarrow E(\omega) = W(\omega) [H_d(e^{j\omega}) - H_d(e^{j\omega})] \quad \dots \text{Error response.}$$

$\hookrightarrow$ weighting function.

$$W(\omega) = \begin{cases} \frac{\delta_s}{\delta_p} & \dots \text{passband.} \\ 1 & \dots \text{stopband.} \end{cases}$$

$$W \in R$$

$$R \triangleq [0, \omega_p] \cup [\omega_s, \pi]$$

$$\Rightarrow E(\omega) = W(\omega) \cdot Q(\omega) \left[\frac{H_d(e^{j\omega})}{Q(\omega)} - P(\omega) \right].$$

$$\Rightarrow E(\omega) = \hat{W}(\omega) \cdot [\hat{H}_d(e^{j\omega}) - P(\omega)]$$

Minimize the Max. error.

$$\{\alpha(k)\} \Rightarrow a(k), \tilde{b}(k), \tilde{c}(k), \tilde{d}(k).$$

$$= \min_{\{\alpha(k)\}} \left\{ \max_{\omega \in R} |E(\omega)| \right\}$$

$\Rightarrow$ minimization, over $\alpha(k)$, of max Error function for frequencies of range R .

$\Rightarrow$ alternation theorem.

Parks - Mclellan Algo.

Remez - exchange Algo.